IIT BOMBAY TECHFEST 2026–27 | NAMMA SPACE COMPETITION

Standard Operating Procedure (SOP): Indoor Spatial Capture

Standardized Protocol for High-Fidelity Monocular Smartphone 3D Gaussian Splatting

1. Pre-Capture Environment Preparation

1. Illumination Consistency: Turn on all overhead artificial lights to provide even, diffuse lighting across the space. Close blinds if direct sunlight casts moving shadows or intense glare. Never switch lights on or off during capture.

2. Static Space Enforcement: The room must be static. Remove pets, turn off ceiling fans, and ensure no moving individuals enter the frame to prevent "ghost" floaters.

3. Featureless Surface Mitigation: On featureless blank white walls, temporarily attach sticky notes, posters, or textured markers to provide anchor points for COLMAP feature matching.

4. Specular Surfaces: Cover full-length mirrors and reflective glass panes to avoid virtual inverted reflections that corrupt depth estimation.

2. Hardware & Camera Configuration

- **Device:** Standard Smartphone (iOS or Android)
- **Resolution & Frame Rate:** 4K UHD (3840×2160) preferred; minimum 1080p FHD at 30/60 FPS.
- **Lens Selection:** Primary Wide lens (1x). Avoid ultra-wide/fisheye (0.5x) to eliminate optical barrel distortion.
- **Exposure / Focus Lock (Mandatory):** Point the camera at a medium-lit spot and long-press to activate **AE/AF Lock**, preventing exposure breathing during pans.

3. The 4-Pass Orbital Trajectory

To guarantee the mandatory >75% frame overlap, capture follows a continuous four-pass path:

1. **Pass 1 — Eye-Level Perimeter Orbit (0° Pitch):** Walk slowly around the outer room perimeter at eye height (~1.6 m) facing slightly inward to establish horizontal perimeter anchors. Maintain steady heel-to-toe walking.
2. **Pass 2 — Low-Angle Floor & Furniture Orbit (+25° Pitch):** Walk at chest level (~1.0 m) angled down toward the floor, desk surfaces, and chair legs to anchor ground plane geometry.
3. **Pass 3 — High-Angle Ceiling Orbit (−25° Pitch):** Walk with phone raised (~1.8 m) angled toward ceiling light fixtures, wall decor, and upper corners to capture vertical bounds.
4. **Pass 4 — Center Inward Sweep:** Step to the room center, perform a slow 360° pan, and sweep key Points of Interest (workstation, display, doorway) for micro-detail feature density.

4. Extraction & Reconstruction Pipeline

- **Frame Extraction:** Extract frames at **7 to 10 FPS** via FFmpeg (`ffmpeg -i input.mp4 -vf "fps=7,scale=1920:-1" frames/%05d.png`), yielding 180–220 dense frames per 30-second capture.
- **Reconstruction Parameters:** Headless COLMAP using the `SIMPLE_RADIAL` camera model.
- **Training Iterations:** 7,000 steps minimum on NVIDIA GPU, achieving >50 dB PSNR.

- **Deployment Packaging:** Export as compressed `.splat` binary (715 KB) for instant zero-install WebGL streaming at 60+ FPS.